

1.4.1 “Where did my page go?”—How Virtual Memory Works

Consider the case of a process running on your system. It's one that requires a large amount of memory, but your OS cannot spare that much. What it does instead is load only the most essential part of the process into the physical memory, putting the rest in the virtual memory. Soon enough, the process will raise a 'page fault', because it couldn't find a page it wanted in the physical memory. This, of course, is because your OS put it on the hard drive. At this point, the OS suspends this process, and begins to look for the page in the virtual memory. If it cannot find the page, it tells you that the process cannot go on under these conditions, and kills it. If, however, it does find the required page, it begins to look for free

Memory Pages

Imagine you are looking for a word in the dictionary. You open the dictionary, turn to the page that has your word, and move your finger down to hunt for it. Memory is organised in much the same way. The entire system RAM is your dictionary, memory is organized into pages which hold lots of data, and the OS runs its supposed 'finger' down the page to find the data it is looking for.

space in the physical memory for it to store this page. If the memory is full, the OS will look for pages that haven't been used in a long time; it will then take the oldest of these pages, put it into the virtual memory, and use the newly freed space to load the page it got from the virtual memory. It now wakes up the process and tells it that its page is ready.

1.5 Putting It All Away—Managing Storage

Your data is organized into *files*—collections of data that represent one single unit. All operating systems have a preferred method to organise these files on storage devices, to make them easy to find and use.

Before we look at how your operating system organises these files, let's take a look at what it's organising them on. Most storage media is in the form of a disk, organized into tracks, sectors and blocks. Sectors are like slices of a pizza—any disk is made up of a large number of sec-